

Stream API

exercises

Task 1. Find the longest string

Given a list of string find the longest one using stream API

Task 2. Sort list in ascending/descending

Sort a list of numbers in ascending and then descending order.

Task 3. Given a list of lists of integers. Your goal is to:

1. Flatten it into a single list.
2. Remove duplicate values.
3. Filter only numbers greater than 10.
4. Sort the final list in ascending order.

Task 4. Given a list of strings. You need to:

1. Split all sentences into words.
2. Count how many times each word appears.
3. Use parallel stream to improve performance on large datasets.

Task 5. Exploring various functions in stream api

```
List<String> names = Arrays.asList("Alice", "Bob", "Charlie", "David", "Eve", "Bob", "Alice", "Frank");
```

Consider above lists and perform –

- Convert all names to **uppercase**.
- Count how many names start with the letter 'A'.
- Group names by their **length** (e.g., names of length 3, 4, etc.).
- Create a **comma-separated string** of all names.
- Use `peek()` to print each name while processing.
- Run any operation using **parallel stream** and show thread names via `peek()`

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